IKBEL HAMDI

Engineering Student — Software Development, Data & AI, DevOps

Full-Stack & Web · Desktop · GenAI & Agents · Machine Learning & Vision · MLOps/DevOps

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PROFILE

Final-year engineering student, passionate about designing complete and intelligent digital solutions. Strong skills in **software development** (web, desktop), **artificial intelligence** and **data science**, **data analysis** and **Business Intelligence**, as well as **DevOps** and **MLOps** to automate and secure the application lifecycle. Autonomous, rigorous and comfortable in a team, I turn business needs into concrete products, from data to production. **Seeking a 6-month end-of-studies internship starting January 2027.**

TECHNICAL SKILLS

Software Development	Python, Java (JavaFX, Spring Boot), C, C++ (Qt), C#/.NET, PHP/Symfony, HTML/CSS/JavaScript, React, FastAPI, REST APIs, Streamlit, Arduino
Web & Databases	Symfony, JWT/OAuth, Stripe, MySQL, PostgreSQL, Oracle, MongoDB, SQL, ChromaDB, FAISS
GenAI & Agents	LLMs (Gemini, Ollama, Qwen), RAG, LangChain, LangGraph, Hugging Face, BGE embeddings, multi-agent systems, Prompt Engineering, speech-to-text
ML / Deep Learning	Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, SVM, ANN, RNN, LSTM, GRU, Transformers, PyTorch, TensorFlow/Keras, time series (R, ARIMA)
Computer Vision	OpenCV, YOLOv8, MediaPipe, MiDaS, MobileNetV2, EfficientNet, ResNet18, EasyOCR, BLIP, Grad-CAM (XAI)
MLOps & DevOps	CI/CD, Git/GitHub, MLOps, MLflow, n8n (workflow automation), testing (pytest)
Data & BI	Pandas, NumPy, Power BI, Excel, Plotly, Matplotlib, Seaborn, Big Data

PROFESSIONAL EXPERIENCE

AI Engineer Intern — RAG & Generative AI · Sotetel (Telecom) — Jun 2026 – Aug 2026

- RAG assistant for telecom technical documentation: extraction and structuring pipeline for unstructured PDFs, semantic search (LangChain, ChromaDB) and local LLM inference (Ollama).

Software & AI Developer Intern · CTN Tunis — 1.5 months

- Complaint management application: generative AI (Gemini), voice-to-text input, e-mail notifications and Oracle database.

PROJECTS

OptiRH — HR Management, Web & Desktop · Symfony, JavaFX, MySQL — Team project

- Complete HR solution, web (Symfony) and desktop (JavaFX): secure accounts and roles, job and internship offers, events (Stripe payment, QR tickets, SMS, maps, weather), AI-analyzed complaints, transport and mission tracking.

AzulTunes — Tourism Web Platform · HTML/CSS/JS, AI chatbots — Top 10 ESPRIT 2024

- Web platform promoting tourism in Tunisia: secure users, events and QR tickets, hotel and restaurant offers with recommendations, collaborations, blogs, complaints, AI chatbot, e-mail and SMS.

BCMID — Multimodal Breast Cancer Diagnosis with XAI · PyTorch, FastAPI, FAISS, n8n — Team project

- Multi-agent diagnostic support system (mammography + ultrasound): anonymization, ResNet18 + Grad-CAM, BGE/FAISS RAG, safety agent, FastAPI API and end-to-end n8n orchestration.

Multi-Agent BI System · LLM Agents, KPI — Ongoing

- Specialized agents (data cleaning, data engineering, BI/KPI reporting) automatically generating dashboards from multiple sources.

AgriVAI · Project Lead · IoT, ML, DL, LLM — Team project

- Intelligent agritech platform (IoT, ML, DL, LLM) with six modules (irrigation, fertilization, crops, diseases, livestock); best models: LightGBM (88%), CatBoost (R^2 = 0.99), Random Forest (96%), MobileNetV2 (96%), EfficientNetB0 (93%).

Other projects

- **Assistive Vision**: YOLOv8n, EasyOCR, BLIP and real-time voice feedback (Streamlit). **GemTrack Pro**: C++/Qt desktop application with Arduino.
- **FIFA World Cup 2026**: match prediction (Gradient Boosting, football-data.org API). **OncoRisk AI**: cancer risk from medical imaging, 8 models (98%).
- **Delivery Prediction**: 150,000+ couriers, RF (MAE 6), K-Means. **2E-EVRP**: ant colony optimization. **Sign Language**: MLP 99% / LSTM 75%.

EDUCATION

Engineering Degree in Data Science & AI · ESPRIT, Tunisia (final year) — 2022 – 2027

Baccalaureate, Experimental Sciences — 2022

CERTIFICATIONS & LANGUAGES

NVIDIA DLI: Deep Learning, NLP, Generative AI, LLM applications, anomaly detection | **MASSAI Summer School (2025):** Generative AI, LLMs & AI agents | **365 Data Science:** SQL (2025)

Languages: Arabic (native) · French C1 · English C1 · Turkish A2